

Latest Innovations In Power Management ICs

With increasing product specifications, power management solutions are also upgrading. The article discusses major innovations seen in power management ICs in last one year

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Power management plays a critical role in terms of efficiency of the device, be it an automotive high-voltage power converter or an ultra-low-power wireless sensor network. With an increase in demand for proper power management, many companies are looking at manufacturing customised integrated chips (ICs). Let us have a look at some new features that have emerged in the power management integrated chips (PMICs) launched in last one year.

Multiple outputs

Recently-launched power management solutions feature multiple outputs employing DC-DC converters and low dropouts (LDOs). Industrial applications require multi-channel outputs for multiple distinct peripherals in a system. For example, a battery-powered IoT system might need multiple voltage rails of different voltage levels for powering sensors, onboard

MCUs, voltage regulators, battery charger ICs, etc. Some power rails may require high voltage accuracy and low noise, like voltage references. This high-accuracy voltage is usually supplied by low-noise, high-PSRR (power supply rejection ratio) LDOs integrated within the PMICs.

Maxim Integrated's MAX2520 features three DC-DC converters and a high-PSRR LDO intended for camera applications. The first output is from a high-voltage step-down converter, which steps down the voltage from a car battery. The other two outputs are from a low-voltage step-down converter, and the fourth output is from a high-PSRR LDO for powering the imager.

NXP Semiconductor's VR5510 PMIC built for infotainment applications has three low-voltage buck converters, a boost converter, three LDOs, and a high-voltage LDO.

Texas Instruments (TI) TPS65921 integrated power management system, designed to power peripherals of multimedia processors, features three step-down converters and four general-purpose LDOs. TI's TPS65951, built for mobile phones, features three step-down converters and ten external LDOs for clocks and peripherals.

Programmable output voltage

Adjustable outputs in PMICs provide flexibility to design engineers in creating their own integrated system. Customised devices can benefit a lot from PMICs' multiple programmable outputs. Integrated switching FETs or software control the output voltages depending on the product.

TI's TPS658643 advanced power management IC is designed for handheld

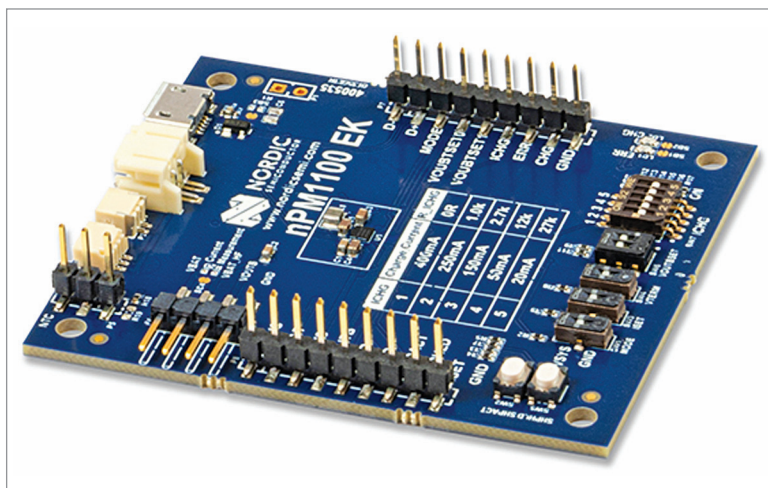


Fig. 1: Nordic Semiconductors' nPM1100 (Credit: <https://www.nordicsemi.com>)